

# Optimization

## ▼ What is optimization?

Optimization is a technique or a mathematical application through which we can maximize or minimize a function. The function may be of one or several variables and when we are optimizing it, that is when we are maximizing or minimizing the function that may have subject to certain constraints (limitations).

## ▼ The goal of optimization technique

How to maximize or minimize (based on what data is available. i.e to measure profit = maximize and for cost = minimize) a function subject to certain constraints

### Review

 <https://www.studocu.com/row/document/university-of-alberta/mathematical-programming-and-optimization-i/review/15253447>

**Objective function:** Function to be minimized (cost).  
**Constraints:** Linear inequalities or equalities on decision variables.  
**Feasible solution:** Decision variables that satisfy the constraints.  
**Feasible set:** Set of all feasible solutions.  
**Optimal value:** Desired minimum.  
**Optimal solution:** A feasible solution that achieves the optimal cost.  
**Optimal set:** Set of all optimal solutions.  
**Cost vector:** Vector of cost weights for each decision variable.  
**Linear:**  
**Standard form:**

$$\text{minimize } c^T x$$

### Mathematical optimization

lessons and solutions

 <https://www.studocu.com/row/document/university-of-baguio/differential-calculus/mathematical-optimization/37776377>

that  $f(x) \geq f(x)$  for all  $x \in A$  ("maximization")  
 Such a formulation is called an **optimization problem** or a **mathematical programming problem** (a term not directly related to **computer programming**, but still in use for example in **linear programming** – see [link](#) below). Many real-world and theoretical problems may be modeled in this general framework.  
 Since the following is valid  

$$f(x) \geq f(x) \Leftrightarrow f(x) \leq f(x)$$
  
 with  

$$f(x) = -f(x), f: A \rightarrow \mathbb{R}$$
  
**Maxima and Minima**  
**mathematical analysis**, the **maxima and minima** (the respective plurals of **maximum** and **minimum**) of a **function**, known collectively as **extrema** (the plural of **extremum**), are the largest and smallest value of the function, either within a given range (the **local** or **relative** extrema) or on the entire **domain of a function** (the **global** or **absolute** extrema). [read](#) [Pierre](#)

<https://web.stanford.edu/group/sisl/k12/optimization/MO-unit1-pdfs/1.1optimization.pdf>

### A Gentle Introduction to Optimization / Mathematical Programming - MachineLearningMastery.com

Whether it is a supervised learning problem or an unsupervised problem, there will be some optimization algorithm working in the background. Almost any classification, regression or clustering problem can be cast as an optimization problem. In this tutorial, you will discover what is optimization and concepts related to it. After

 <https://machinelearningmastery.com/a-gentle-introduction-to-optimization-mathematical-programming/>



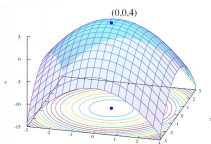
### Mathematical optimization | Engati

Mathematical optimization is the selection of a best element, with regard to some criterion, from some set of available alternatives

 <https://www.engati.com/glossary/mathematical-optimization>

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What is  
**Mathematical  
Optimization?**



### 2019 0709 mathematical optimization

無料の要約、過去の試験、講義ノート、解決策などを共有しましょう。

 <https://www.studocu.com/row/document/東北大学/applied-data-science/2019-0709-mathematical-optimization/6784670>

20190709 - 20190723 Applied Data Science  
**Machine learning introduction**

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